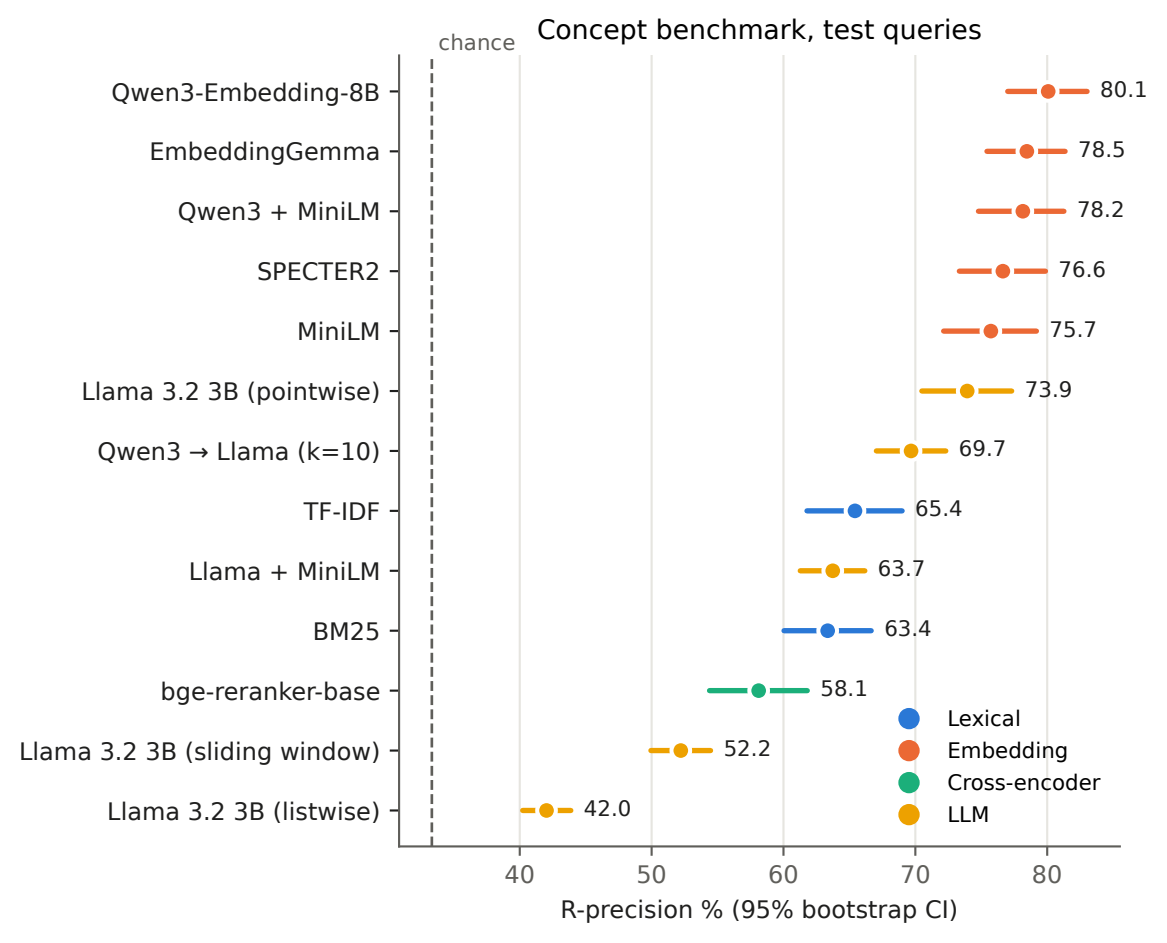
Concept benchmark, test queries
chance
Qwen3-Embedding-8B
EmbeddingGemma
Qwen3 + MiniLM
SPECTER2
MiniLM
Llama 3.2 3B (pointwise)
Qwen3 → Llama (k=10)
TF-IDF
Llama + MiniLM
BM25
bge-reranker-base
Llama 3.2 3B (sliding window)
Llama 3.2 3B (listwise)
80.1
78.5
78.2
76.6
75.7
73.9
69.7
65.4
63.7
63.4
58.1
52.2
42.0
Lexical
Embedding
Cross-encoder
LLM
40
50
60
70
80
R-precision % (95% bootstrap CI)